Ablation Study of the Sift Pipeline

Which components of the alignment cross-encoder matter: inputs, consolidation, loss, heads, denoising and compression

PROJECT	DELIVERABLE	TEAM NO	TEAM MEMBER	TEST COHORT
Sift	**Deliverable 5: Ablation Study**	**Team 24 (single member)**	**Kundhave S, CB.SC.U4CSE23037**	**2,700 hunks (httpx, attrs, axios)**

1. Executive Summary and Study Objectives

The ablation isolates each design choice of the Sift pipeline by removing it and re-training under the same protocol, then evaluating on the same held-out repository-disjoint test split. The headline finding is that removing the hunk or the intent is clearly harmful, but removing consolidation, focal loss, the multi-task heads or denoising did **not** lower the score in these single-seed runs. The study therefore does not show that those four components help; they may still matter on other data or seeds, but this experiment gives no evidence for them.

- Full model: macro-F1 0.680, NDCG@5 0.868, hard-tier pairwise AUC 0.726, 596 MB.
- Deployed target (INT8 ONNX student): macro-F1 0.667, NDCG@5 0.869, hard-tier AUC 0.681, 22.9 MB, CPU p50 / p95 8.5 / 66.9 ms.

2. Experimental Methodology and Evaluation Cohort

Cohort. The test split of Sift-IA v1.0: 2,700 hunk-intent pairs from three repositories (httpx, attrs, axios) that appear in no training or validation row; 594 are misaligned. Labels are evaluated as *label_clean*. Rankable sessions (at least 2 hunks and 1 misaligned) number 456 in the stored evaluation. **Training.** All variants use the full main-model protocol (ModernBERT-base, L = 512, micro-batch 8 x 2, bf16, AdamW 2e-5, up to 6 epochs, early stopping on validation macro-F1) on the same rule and confident-learning denoised rows, except the no-denoising run. **Decision threshold** is tuned on validation macro-F1 only. **Metrics:** accuracy, macro-F1, per-class F1, precision, recall, specificity, within-session pairwise AUC per difficulty tier, NDCG@5, model size and CPU latency. **Seeds:** single seed (473), no confidence intervals; single seed (473) for every row: differences of about 0.01 macro-F1 are within the run-to-run noise one should expect and are not significant.

3. Description of Configurations

#	Configuration	What is changed	Status
1	Full model	ModernBERT cross-encoder, consolidated intent, focal loss, three heads, denoised rows	run
2	Hunk-only	Intent text removed from the input	run
3	Intent-only	Hunk text removed from the input	run
4	No consolidation	Raw multi-turn intent text, including chit-chat and retracted turns	run
5	No focal loss	Alignment term is plain cross-entropy	run
6	Alignment head only	Drift-type and confidence heads and their loss terms removed	run
7	No denoising	Train on all 12,600 rows with the noisy labels as shipped	run
8	Distilled MiniLM student	22M-parameter student, tau 4, alpha 0.5; extra rows: no distillation, ONNX fp32	run
9	INT8 ONNX student	Dynamic INT8 quantisation of the student (deployed target)	run
-	Rules-only denoising	Rule filters without confident learning (experiments/run_all_gpu.sh, abl_rules_only)	-
-	K-means features on / off	Cluster prior and centroid distance added to a Platt calibrator (results/kmeans.json)	run, small study

4. Results

Table 1: classification at the validation-tuned threshold (test split, label_clean). Per-class F1 comes from the stored eval JSONs (computed by experiments/evaluate.py from each run's predictions).

#	Configuration	Acc.	Macro-F1	F1 aligned	F1 misal.	Precision	Recall	Specif.
1	Full model (baseline)	0.787	0.680	0.865	0.496	0.516	0.476	0.874
2	Hunk-only input (no intent)	0.497	0.462	0.599	0.326	0.231	0.552	0.481

#	Configuration	Acc.	Macro-F1	F1 aligned	F1 misal.	Precision	Recall	Specif.
3	Intent-only input (no hunk)	0.701	0.548	0.811	0.284	0.300	0.269	0.823
4	No intent consolidation (raw turns)	0.792	0.689	0.868	0.510	0.529	0.492	0.877
5	No focal loss (plain CE)	0.810	0.696	0.882	0.510	0.591	0.448	0.913
6	Alignment head only (no multi-task)	0.812	0.706	0.883	0.529	0.590	0.480	0.906
7	No denoising (noisy labels as is)	0.773	0.688	0.851	0.525	0.486	0.571	0.830
8	Distilled MiniLM student (fp32, PyTorch)	0.787	0.667	0.867	0.467	0.521	0.423	0.890
8b	MiniLM student, no distillation (extra)	0.768	0.634	0.855	0.412	0.465	0.370	0.880
8c	Student ONNX fp32 (extra)	0.774	0.669	0.856	0.483	0.487	0.478	0.858
9	INT8 ONNX student (deployed target)	0.779	0.667	0.860	0.474	0.497	0.453	0.871

Table 2: ranking quality per difficulty tier, model size and single-pair CPU latency (p50 / p95, ms).

#	Configuration	NDCG@5	AUC random	AUC hard	Macro-F1 random	Macro-F1 hard	Size MB	CPU p50 / p95 ms
1	Full model	0.868	0.800	0.726	0.717	0.632	596.1	94.5 / 95.1 *
2	Hunk-only input	0.780	0.561	0.592	0.512	0.461	596.1	94.5 / 95.1 *
3	Intent-only input	0.726	0.505	0.501	0.543	0.542	596.1	94.5 / 95.1 *
4	No intent consolidation	0.891	0.831	0.738	0.726	0.645	596.1	94.5 / 95.1 *
5	No focal loss	0.887	0.844	0.712	0.727	0.640	596.1	94.5 / 95.1 *
6	Alignment head only	0.895	0.857	0.752	0.742	0.653	596.1	94.5 / 95.1 *
7	No denoising	0.902	0.861	0.767	0.739	0.666	596.1	94.5 / 95.1 *
8	Distilled MiniLM student	0.869	0.794	0.678	0.696	0.621	90.9	-
8b	MiniLM student, no distillation	0.842	0.759	0.632	0.653	0.587	90.9	-
8c	Student ONNX fp32	0.874	0.794	0.695	-	-	90.4	13.4 / 68.7
9	INT8 ONNX student	0.869	0.784	0.681	-	-	22.9	8.5 / 66.9

* Same architecture and size as configuration 1: the latency of the full model (94.5 / 95.1 ms, fp32 PyTorch, L = 512, 14 threads, idle machine, i7-13650HX) is shown for configurations 2-7 and was not timed separately. Student latencies were measured with 4 threads while a training job shared the CPU, so they are not strictly comparable with the full-model figure. PyTorch student latency was not measured; INT8 and ONNX fp32 were. AUC and Macro-F1 per tier: tiers keep sessions that contain a misaligned hunk of that tier.

K-means features (results/kmeans.json): Platt on the logit alone gives macro-F1 0.680; adding cluster features gives 0.691 (+0.010), a change within single-seed noise.

5. Insights

- **Inputs matter:** hunk-only macro-F1 0.462 (-0.218 vs full) and intent-only 0.548 (-0.133); intent-only hard-tier AUC is 0.501, close to chance, so the pair is needed and the model is not just reading the intent.
- **Components not shown to help:** no intent consolidation macro-F1 0.689 (+0.009); no focal loss macro-F1 0.696 (+0.016); alignment head only macro-F1 0.706 (+0.026); no denoising macro-F1 0.688 (+0.007). All four ablated variants scored higher macro-F1 than the full model, by 0.007 to 0.026. With one seed and differences of about 0.01 in noise, the honest reading is that these components have no measurable benefit here, not that they hurt.
- **Ranking tells a slightly different story:** hard-tier AUC is higher without denoising (0.767) and with alignment head only (0.752) than for the full model (0.726); on NDCG@5 the full model (0.868) is within 0.034 of every ablated variant.
- **Compression costs little in macro-F1 and NDCG@5 but more on the hard tier:** the INT8 student keeps NDCG@5 (0.869 vs 0.868) and macro-F1 (0.667 vs 0.680) while hard-tier AUC falls from 0.726 to 0.681. INT8 quantisation itself changed macro-F1 by -0.002 against the ONNX fp32 student.
- **Distillation helps the student:** macro-F1 0.667 with distillation vs 0.634 without (+0.033); hard-tier AUC 0.678 vs 0.632.
- **Error bars:** each configuration is a single seed (473), so differences carry no error bar.

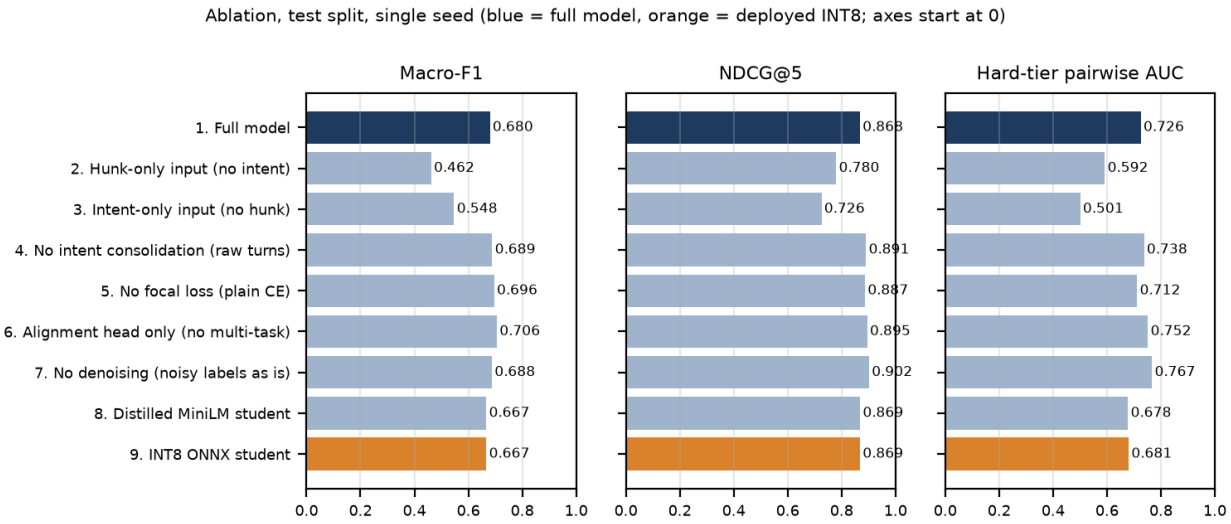


Figure 1. Macro-F1, NDCG@5 and hard-tier AUC for configurations 1-9 (single seed).

6. Deployment Recommendation

Deploy the INT8 ONNX student as the default local verifier. It is the smallest measured configuration (22.9 MB, versus 596.1 MB for the full model, a factor of 26) and meets the planned targets of under 30 MB and p95 under 100 ms on this machine (p95 66.9 ms; targets, not guarantees on other hardware). Its macro-F1 (0.667) and NDCG@5 (0.869) match the full model within 0.013 and 0.000. The cost is on the hard tier (pairwise AUC 0.681 vs 0.726), so where hard-tier ranking matters and a GPU is available the full model is the better choice; it also just meets the CPU target (p95 95.1 ms) but misses the size target.

Because the four ablated components gave no measurable gain, a simpler training recipe (cross-entropy, alignment head only, no consolidation, no denoising) is a defensible alternative that this study cannot rule out; it would need repeated seeds to decide. Command to repeat all ablation runs on a GPU machine: `bash run_gpu_experiments.sh`.

7. Scope and Limitations

- Labels in Sift-IA v1.0 are construction-determined (a misaligned hunk is a donor hunk from a different commit); there is no human validation.
- The test split (httpx, attrs, axios) is repository-disjoint but is still open-source Python/JavaScript (it contains no TypeScript); results may not transfer to agent-authored diffs.
- Misaligned hunks are constructed, so a model could learn construction artefacts; the hard tier mitigates this but does not remove it.
- Latency and size come only from the machine they were measured on (Intel i7-13650HX CPU, RTX 4050 Laptop GPU with 6 GB); every run is a single seed (473), no confidence intervals.
- Baselines, noise-robustness and hyper-parameter search use reduced settings; the rules-only denoising ablation, IG convergence checks, a user study of explanations, and drift and bias measurement are outside this work.